(b)	Write a complete program to implement a circular queue based on an array as element container. Program should have functions for insertion, deletion , getLength and traversal from front to rear (i.e. display queue elements).	6
(c)	Consider a linked implementation of a binary tree. Binary tree node structure is defined by BTNODE structure. Also consider you are having a queue implementation to store BTNODE pointer type of values. Given root pointer to a binary tree write C/C++ function to implement level order traversal of the tree.	6
Q4.(a)	Consider an empty binary search tree (which stores character values). Following characters are inserted in given sequence in the tree : d h k r y a p b m z c Draw the tree after all values inserted and write inorder and postorder traversal sequence of the tree.	3
(b)	Define AVL tree? Justify the need of height balancing in the binary search tree. Illustrate LL and RL rotations.	3
(c)	Differentiate M-way search tree and B tree.	3
(d)	Given the root pointer of a binary tree of integer values (implemented as linked nodes), write a C/C++ function to compute sum of all nodes' values.	3
(e)	What is the basic motivation to use the m-way search tree instead of binary search tree? Given following values to be inserted in a 5-way B-tree: 12, 7, 56, 2, 13, 44, 9, 19, 20, 10, 5, 22, 13 Illustrate the B-tree after all values inserted.	3
Q5.(a)	Briefly explain and compare adjacency matrix, incidence matrix and adjacency list representation approach for memory representation of graph. How much memory shall be required in each representation to store a directed unweighted graph having N number of vertex and M number of edges?	6
(b)	Explain traversal methods in graphs. Write algorithm for breadth first traversal.	6
(c)	Consider an undirected weighted graph represented by the following adjacency matrix. Compute shortest path cost to all nodes from source node **a**. # indicates no edge.	6

	a	b	c	d	e	f
a	0	1	#	8	#	#
b	1	0	6	#	#	2
c	#	6	0	#	7	#
d	8	#	#	0	#	2
e	#	#	7	#	0	1
f	#	2	#	2	1	0